4th-Order Butterworth Low-Pass: Sallen-Key Derivation Notes

Specification: fc = 10 kHz, two cascaded unity-gain Sallen-Key stages, base resistor R = 10 kΩ.

1. Circuit and assumptions

One stage: R_1 connects the input to node X. R_2 connects X to the op-amp + input (node P). C_1 connects X to the output V_o. C_2 connects P to ground. The op-amp is a unity-gain buffer (the − input is tied to the output).

Feedback check: the − input is fed straight from the output, so $\beta_{neg} = 1$. The only positive feedback is through C_1, which gives $\beta_{pos} < 1$ (it depends on frequency). Negative feedback dominates, so the circuit is stable and the virtual short applies.

$$V^- = V^+ \quad \text{and} \quad V^- = V_o \quad \Rightarrow \quad V^+ = V_o$$

> *This is a virtual short ($V^+ = V^-$), not a virtual ground, because the + input is not at 0 V.*

2. Transfer function of one stage

Step 1. No current enters the op-amp, so the current through R_2 flows into C_2:

$$(V_x - V^+) / R_2 = V^+ \cdot sC_2$$

$$V_x = V_o(1 + sR_2C_2) \qquad \ldots(1)$$

Step 2. KCL at node X (currents leaving X through R_2 and C_1):

$$(V_{in} - V_x) / R_1 = (V_x - V_o) \cdot sC_1 + (V_x - V_o) / R_2$$

Step 3. Multiply by R_1:

$$V_{in} - V_x = (V_x - V_o)(sC_1R_1 + R_1/R_2)$$

Step 4. From (1), $V_x - V_o = V_o \cdot sR_2C_2$. Substitute:

$$V_{in} - V_x = V_o \cdot sR_2C_2\,(sC_1R_1 + R_1/R_2)$$

$$= V_o\,(s^2R_1R_2C_1C_2 + sR_1C_2)$$

Step 5. Add V_x from (1):

$$V_{in} = V_o\,[\,(1 + sR_2C_2) + sR_1C_2 + s^2R_1R_2C_1C_2\,]$$

$$\mathbf{H(s) = V_o/V_{in} = 1 \,/\, (\, s^2R_1R_2C_1C_2 + sC_2(R_1 + R_2) + 1\,)}$$

Sanity checks: as $s \to 0$, $H \to 1$ (capacitors open, DC gain 1). As $s \to \infty$, $H \to 0$ (low-pass).

3. Standard form: ω_0 and Q

Divide numerator and denominator by $R_1R_2C_1C_2$ and compare with $s^2 + s\omega_0/Q + \omega_0^2$:

$$\omega_0 = 1 / \sqrt{(R_1R_2C_1C_2)}$$

$$\omega_0/Q = (R_1 + R_2) / (R_1R_2C_1) \quad \Rightarrow \quad Q = \sqrt{(R_1R_2C_1C_2)} / (\,C_2(R_1 + R_2)\,)$$

Special case $R_1 = R_2 = R$:

$$\mathbf{\omega_0 = 1 \,/\, (\, R\sqrt{(C_1C_2)}\,)} \qquad \mathbf{Q = \tfrac{1}{2}\sqrt{(C_1/C_2)}}$$

> *Check: $C_1 = C_2$ gives Q = 0.5 (critically damped, two equal real poles). A larger C_1/C_2 ratio gives a higher Q and more peaking.*

4. Butterworth poles and the Q of each stage

A Butterworth response has all its poles on a circle of radius ω_c in the left half-plane, so both stages use $\omega_0 = \omega_c = 2\pi \times 10$ kHz = 62,832 rad/s. Only Q differs between the stages. For order N = 4 the poles are at 22.5° and 67.5° from the negative real axis, and each conjugate pair forms one second-order stage.

Take θ as the angle from the negative real axis. A pole pair is then:

$$s = -\omega_0 \cos\theta \pm j\omega_0 \sin\theta$$

The standard second-order poles are:

$$s = -\omega_0/(2Q) \pm j\omega_0 \sqrt{(1 - 1/(4Q^2))}$$

Equating the real parts, $\omega_0 \cos\theta = \omega_0/(2Q)$, so:

$$\mathbf{Q = 1 / (2 \cos\theta)}$$

Stage	θ	Q = 1/(2 cos θ)	Pole position
A (first)	22.5°	0.5412	Closer to the real axis: low Q, little peaking
B (second)	67.5°	1.3066	Closer to the jω axis: high Q, more peaking

> *A pole nearer the real axis (small θ) has a larger cos θ and therefore a lower Q. A pole nearer the jω axis (large θ) has a higher Q.*

Cross-check: at $\omega = \omega_0$, each stage has $|H| = Q$ (because $H(j\omega_0) = Q/j$). The cascade therefore has $|H(j\omega_c)| = Q_A \times Q_B = 0.5412 \times 1.3066 = 0.7071 = 1/\sqrt{2}$, which is exactly −3.01 dB at the cutoff, as a Butterworth response requires.

5. Component values for R = 10 kΩ

From $\omega_0 = 1/(R\sqrt{(C_1C_2)})$ and $Q = \tfrac{1}{2}\sqrt{(C_1/C_2)}$ we get $\sqrt{(C_1C_2)} = 1/(\omega_0 R)$ and $C_1/C_2 = 4Q^2$. Solving:

$$\mathbf{C_1 = 2Q / (\omega_0 R)} \qquad \mathbf{C_2 = 1 / (2Q\omega_0 R)}$$

With $\omega_0 R = 62{,}832 \times 10^4 = 6.2832 \times 10^8$:

Stage	Q	C_1 exact	C_2 exact	Check $C_1/C_2 = 4Q^2$
A	0.5412	1.723 nF	1.470 nF	1.172 ($4Q^2 = 1.172$)
B	1.3066	4.159 nF	0.609 nF	6.83 ($4Q^2 = 6.83$)

6. Standard capacitor values and the resulting stage

The nearest E12 capacitors are used. The stage frequency and Q follow from:

$$f_0 = 1 / (2\pi R\sqrt{(C_1C_2)}) \qquad Q = \tfrac{1}{2}\sqrt{(C_1/C_2)}$$

Stage	C_1	C_2	f_0 at R = 10 kΩ	Q
A	1.8 nF	1.5 nF	9.69 kHz	0.548 (target 0.541)
B	3.9 nF	0.56 nF	10.77 kHz	1.319 (target 1.307)

The Q values stay within about 1% of target, because Q depends only on the ratio C_1/C_2. The frequencies drift (stage B by about 8%), because f_0 depends on the product C_1C_2.

7. Tuning the resistors

Q is unaffected by R when $R_1 = R_2$, so f_0 is corrected by changing R alone:

$$R = 1 / (\omega_c \sqrt{(C_1C_2)})$$

Stage	Ideal R	Nearest 1% value	Resulting f_0	Q
A	9.69 kΩ	9.76 kΩ	9.92 kHz	0.548
B	10.77 kΩ	10.7 kΩ	10.06 kHz	1.319

8. Final design

Stage	$R_1 = R_2$	C_1 (feedback)	C_2 (to ground)	Role
A	9.76 kΩ	1.8 nF	1.5 nF	Q ≈ 0.55, first stage
B	10.7 kΩ	3.9 nF	0.56 nF	Q ≈ 1.32, second stage

The cascade (ideal buffers, so the stages do not load each other) multiplies the two transfer functions:

$$H(s) = H_A(s) \cdot H_B(s) = 1 / [(s^2/\omega_A^2 + s/(Q_A\omega_A) + 1)(s^2/\omega_B^2 + s/(Q_B\omega_B) + 1)]$$

This gives an 80 dB/decade roll-off above the cutoff and a phase that reaches −360°. The lower-Q stage is placed first, which is a common ordering.

LTspice results with these values: the −3 dB point is 10,102 Hz with ideal buffers (1% above the 10 kHz target, because of the standard component values) and 10,101 Hz with a real op-amp at a gain-bandwidth of 10 MHz.

9. Quick checks to remember

- $H(0) = 1$, and $H \to 0$ as $s \to \infty$.
- $C_1 = C_2$ gives $Q = 0.5$.
- $Q_A \times Q_B = 1/\sqrt{2} \approx 0.707$ for a 4th-order Butterworth, which gives −3 dB at ω_c.
- $Q = 1/(2 \cos \theta)$, with θ measured from the negative real axis. Small θ gives low Q.
- Q depends on the capacitor ratio; f_0 depends on R and the capacitor product.